

Client / Front-End

POST /reserve

Core Reservation Pipeline

FastAPI Ingestion
Layer
token-bucket rate
limiter
/reserve · /commit
· /release

atomic Lua hold
ack after WAIT

Redis – Hold Engine
+ Event Log
Sentinel: primary +
N replicas
available counter ·
event stream
TTL holds + sweeper

XREADGROUP
micro-batched

Worker Engine
asyncio, N replicas
group-commit
batching

group commit

Postgres – Durable
Ledger
balances ·
reservations
append-only audit
log

periodic diff

Reconciliation
Watchdog
diffs Redis vs.
ledger,
can rebuild on drift

metrics & traces

provisions & deploys

Observability Plane
Prometheus → Grafana
OpenTelemetry ·
Alertmanager

Delivery Plane
CI correctness gate
→ deploy